A MULTIMODAL SYSTEM FOR AUTOMATED CORN DISEASE DETECTION AND FARMER-CENTRIC KNOWLEDGE SHARING

Problem, Its Impact/Gravity

- Corn is highly vulnerable to foliar diseases (e.g., Northern Leaf Blight, Gray Leaf Spot, Common Rust)
- Traditional detection methods are slow, expert-dependent, and not scalable for smallholder farmers
- Delayed interventions lead to:
  - Significant crop losses
  - Reduced farmer income
  - Threats to national food security and sustainability

Who Will Benefit from the Solution

- Smallholder Corn Farmers – especially those with limited access to agronomic expertise
- Agricultural Extension Officers – for efficient farm monitoring and data-driven advice
- Local Agri-businesses & Policy Makers – for better supply planning and support programs
- Research Institutions – for access to real-time field data and collaborative innovation

Presented By: **25-26J-239**
IT4010 Research Project – 2025 July

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A MULTIMODAL SYSTEM FOR AUTOMATED CORN DISEASE DETECTION AND FARMER-CENTRIC KNOWLEDGE SHARING

Design Excellence of the Overall Solution

- Multimodal AI Detection – combines:
  - Computer vision for disease/pest identification
  - Nutrient analysis from tissue samples
- Explainable AI (XAI) – uses Grad-CAM and SHAP for transparent, trustworthy decisions
- Farmer-in-the-Loop Design – integrates farmer feedback to improve model accuracy and adoption
- Unified Decision Support Platform – delivers:
  - Real-time alerts
  - Growth-stage-aware advice
  - Yield predictions
  - Via mobile app in local languages
- Low-cost, scalable, and context-aware – bridges advanced AI with farmer-friendly tools

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CORN LEAF DISEASE DETECTION

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CORN LEAF DISEASE DETECTION

How the solution addresses the sub-problem

- Farmers in Sri Lanka lack low-cost, transparent tools to identify corn foliar diseases early.
- Many models are disease-specific or trained for only a few classes.
- Developed a CNN based deep learning model to identify multiple corn leaf diseases.
  Northern Leaf Blight
  Grey Leaf Spot
  Common Rust
- Integrated the trained model with a Flutter mobile application to allow:
  Image upload from gallery
  Real-time corn leaf disease detection
  Prediction confidence display

User Requirements Addressed by the Solution

- Farmers can upload leaf images directly from mobile phone's camera or gallery
- System provides:
  Disease name
  Confidence level
- Mobile-first design supports field usage with minimal technical knowledge
- The solution satisfies functional requirements for automated disease detection and non-functional requirements such as usability, responsiveness, and system integration.

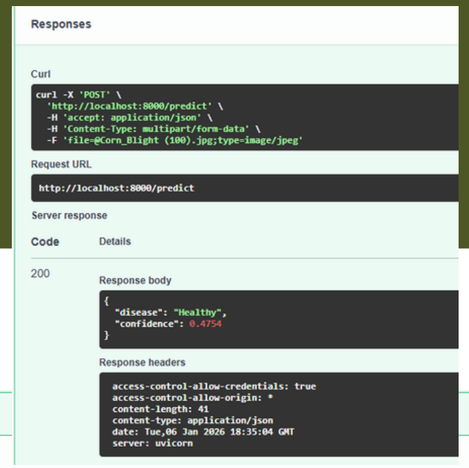
Classification Report:

	precision	recall	f1-score	support
Blight	0.80	0.75	0.77	225
Common_Rust	0.94	0.96	0.95	244
Gray_Leaf_Spot	0.62	0.65	0.64	126
Healthy	0.95	0.97	0.96	242
accuracy			0.86	837
macro avg	0.83	0.83	0.83	837
weighted avg	0.86	0.86	0.86	837

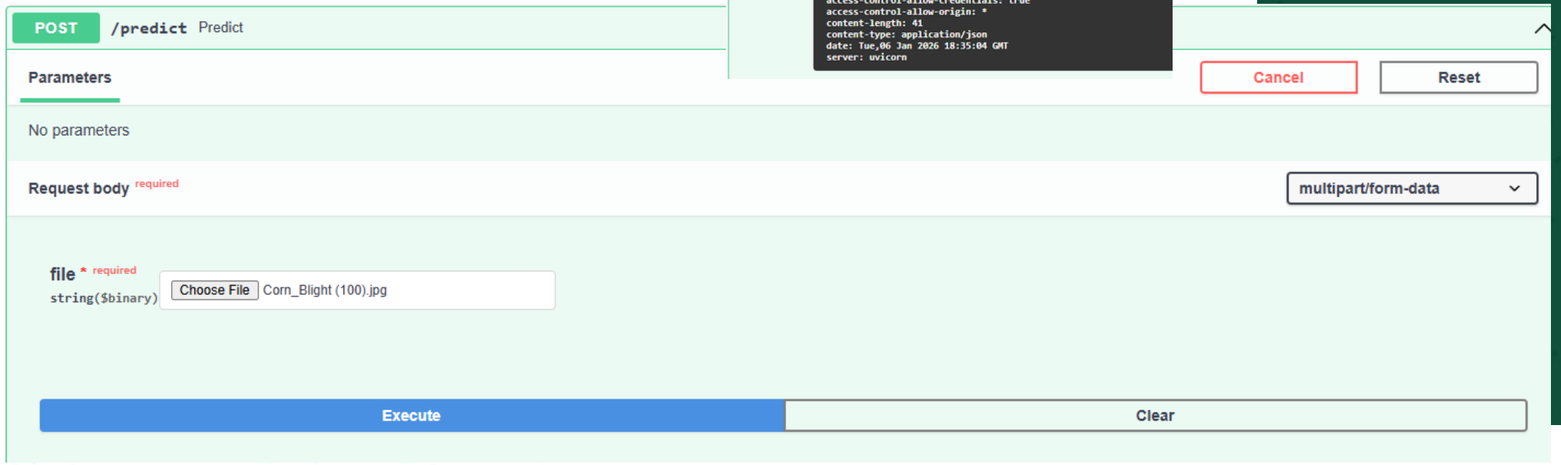
CORN LEAF DISEASE DETECTION

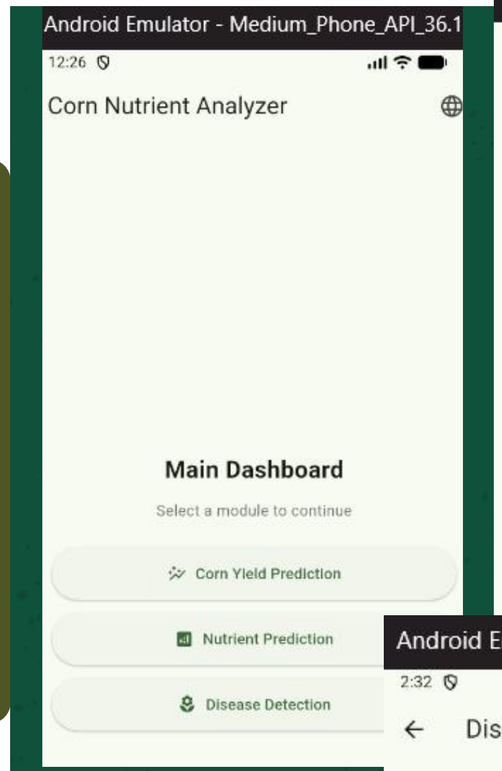
User Feedback on Prototype

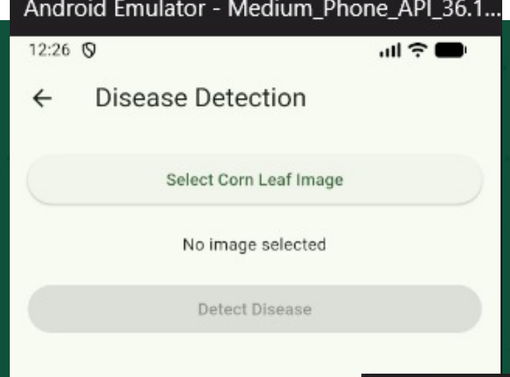
- **Users found the mobile interface:**
- **Simple and easy to navigate**
- **Easy to upload images and understand results**
- **Clear confidence scores helped users understand model reliability**

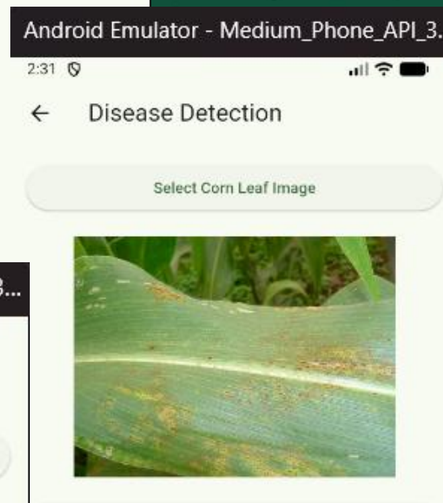


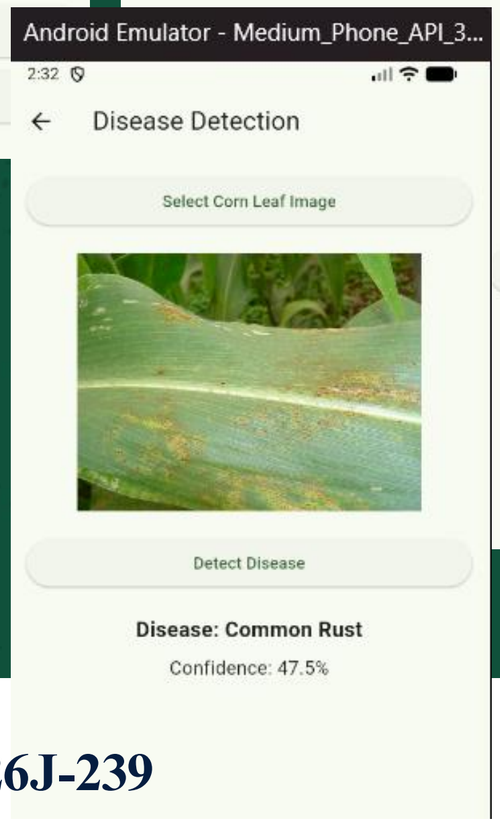
SWAGGER UI











IT22055958 PERERA U S A

PEST DETECTION SYSTEM

Specialization - Information Technology

PEST DETECTION SYSTEM

How the Solution Addresses the Sub-Problem

- Developed a CNN-based deep learning model to detect corn leaf pests:
  - Armyworm
  - Leaf Blight
  - Zonocerus
  - Healthy Corn Leaf
- Model trained using 224×224 RGB images and deployed via FastAPI backend
- Implemented a confidence threshold mechanism to:
  - Detect non-corn images
  - Prevent incorrect predictions on unrelated images
- Integrated the trained model with a Flutter mobile application to allow:
  - Image upload from gallery
  - Real-time pest detection
  - Prediction confidence display

User Requirements Addressed by the Solution

- Farmers can upload leaf images directly from mobile phones
- System provides:
  - Pest type
  - Confidence level
- Invalid images (non-corn leaves) are:
  - Automatically detected
  - Shown a clear warning message
- Mobile-first design supports field usage with minimal technical knowledge

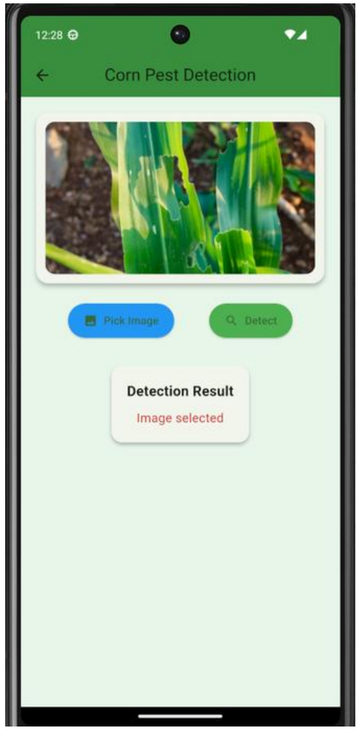
Design Excellence / Contribution

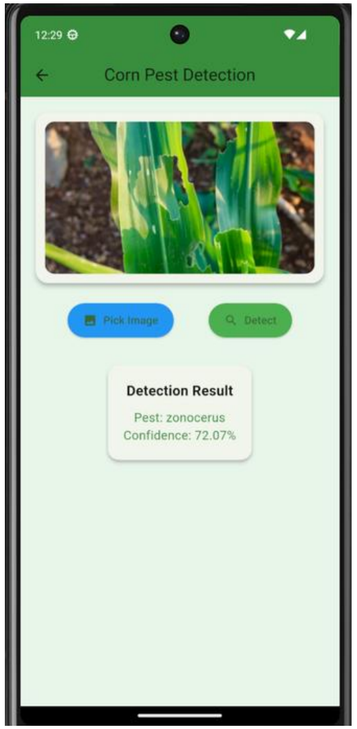
- Model Design
  - Class imbalance handled using class weights
  - Confidence-based rejection of invalid inputs
- Backend Design
  - REST API using FastAPI (/predict)
  - Efficient image preprocessing & inference
- Frontend Design
  - Clean Flutter UI with:
    - Image preview
    - Loading indicators
    - Dialog-based error handling
- Clear separation of:
  - AI logic
  - Backend services
  - Mobile UI

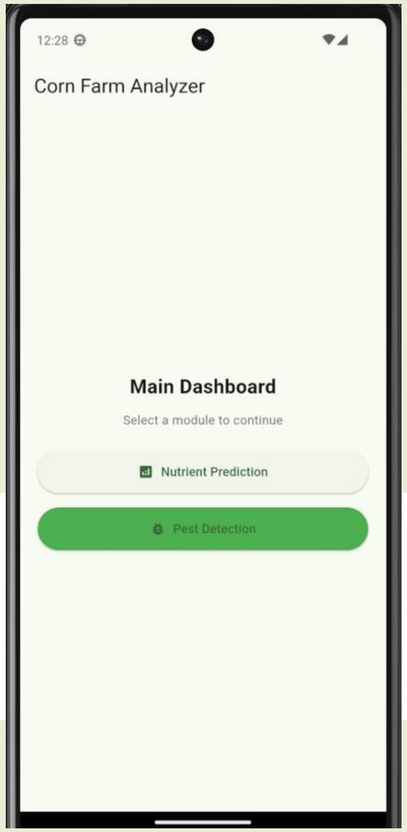
PEST DETECTION SYSTEM

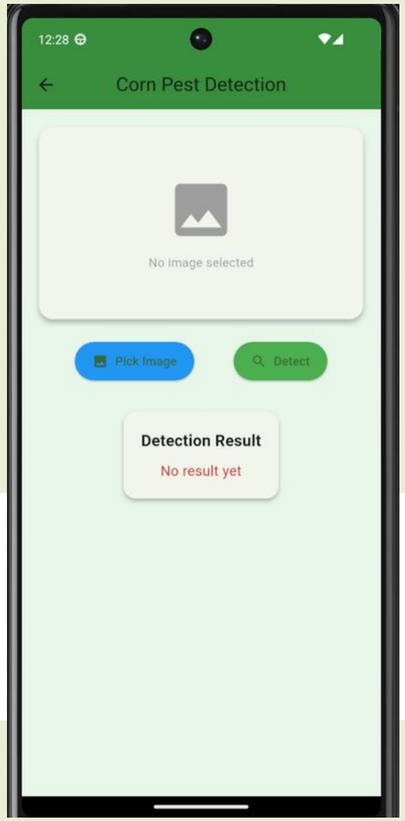
User Feedback on Prototype

- Users found the mobile interface:
  - Simple and easy to navigate
  - Easy to upload images and understand results
- Non-corn image detection improved trust in predictions
- Clear confidence scores helped users understand model reliability

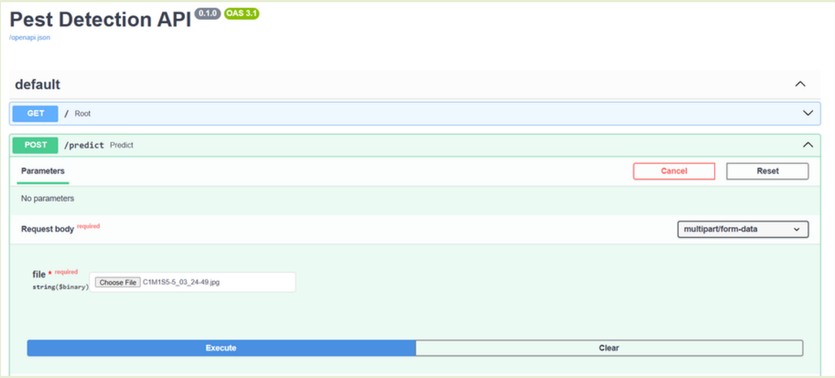


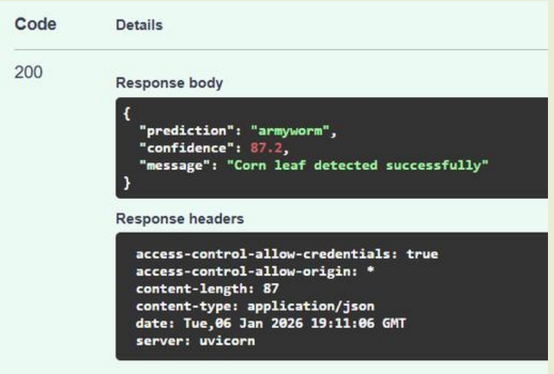






SWAGGER UI





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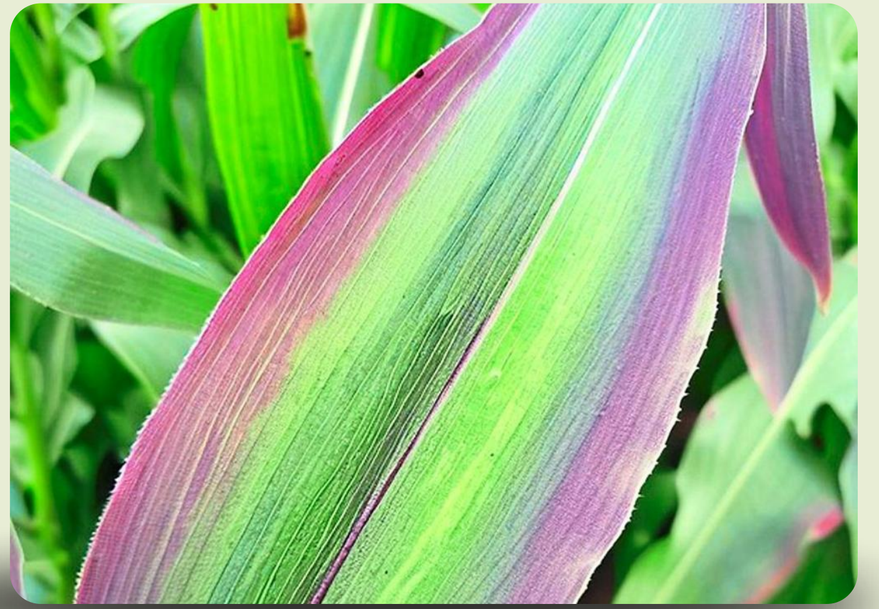
NUTRIENT COMPOSITION PREDICTION

Specialization - Information Technology

NUTRIENT COMPOSITION PREDICTION



Sub Problem Addressed

- Existing nutrient prediction systems focus on single nutrient deficiencies (mainly Nitrogen). Do not consider multi-nutrient interactions within the plant.
- **Traditional soil testing methods** - Time consuming and expensive
- **Visual testing of plant leaves -** Commonly used, but it is not very accurate
- Many nutrient deficiencies show similar visual symptoms.
- Existing systems ignore multiple-nutrient interactions such as P–Zn antagonism or N–K synergy
- Inability to identify nutrient interactions and antagonistic effects is the **key knowledge gap.**
- Farmers receive generic fertilizer advice, leading to imbalance, yield loss, and waste.
- There is no data-driven, field-specific nutrient diagnosis based on tissue samples.
- Comparison of Nutrients with toxicity data and balance the nutrient amounts with standard levels (Processing)

How the Solution Addresses the Sub-Problem

- Uses corn tissue sample data instead of only soil tests for accurate plant nutrient status.
  Consider tissues of the leaves, height, length of the leaves, texture, rolling and curling)
- Applies machine learning models (ResNet -50) to predict nutrient sufficiency/deficiency.
- Performs multivariate analysis (PCA & correlation) to identify nutrient interactions.
- Generates root-cause-based recommendations instead of generic fertilizer advice.

NUTRIENT COMPOSITION PREDICTION

User Requirements Addressed

Farmers can,

- Understand reasons for the deficiency (not just what).
- Receive field-**specific fertilizer recommendations.**
- Reduce unnecessary fertilizer usage.

Researchers can,

- Analyze nutrient trends and interactions.
- Study multi-nutrient behavior across growth stages

- Data Science & ML driven design for precision agriculture.
- Uses:

1. PCA for nutrient interaction discovery
2. ResNet -50 for robust prediction
3. CNN for spectral data (future-ready)

- Integrated Explainable AI (feature importance / SHAP) to improve farmer trust.
- Modular design → easily scalable to other crops

User Feedback on Prototype

- Clear nutrient status (Low / Optimal / Excess).
- Ease of understanding
- Easy navigation
- Simple actionable **advice** instead of scientific jargon.
- Practical fertilizer recommendations

Expected impact:

- 20–30% fertilizer savings
- Improved yield and soil health

```
Classification Report:
              precision    recall  f1-score   support

     Healthy       0.65      0.66      0.65       174
         KAB       0.92      1.00      0.96       516
         NAB       0.56      0.79      0.65       184
         PAB       0.89      0.67      0.76       190
        ZNAB       0.85      0.63      0.72       305

    accuracy                           0.80      1369
   macro avg       0.77      0.75      0.75      1369
weighted avg       0.82      0.80      0.80      1369
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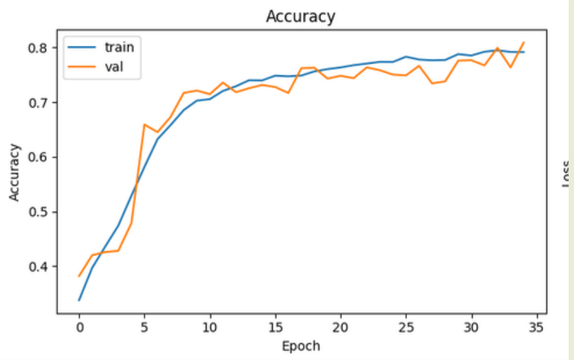

ML-BASED CORN YIELD PREDICTION FOR SRI LANKAN FARMERS

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ML-BASED CORN YIELD PREDICTION FOR SRI LANKAN FARMERS

How the solution addresses the sub-problem

- Sri Lankan smallholder farmers lack accurate and timely methods to estimate corn yield before harvest
- Traditional yield estimation relies on manual surveys and historical averages, which are:
  - Time-consuming
  - Prone to error
  - Not field-specific
- Environmental, agronomic, and farm-level factors affect yield in complex, non-linear ways
- Developed a machine learning–based yield prediction model that:
  - Uses soil, rainfall, fertilizer, pest/disease level, farm size, and previous yield
  - Predicts expected corn yield in kg/acre
- Integrated Explainable AI (SHAP) to highlight:
  - Factors that increase yield
  - Factors that reduce yield

User Requirements Addressed by the Solution

- Farmers can enter field details via a simple mobile interface
- System provides:
  - Predicted yield value
  - Clear breakdown of main contributing factors
- Visual indicators (increases yield / reduces yield) improve understanding
- Mobile-first design supports field usage with minimal technical knowledge
- Solution satisfies functional and non-functional requirements such as:
  - Usability
  - Responsiveness
  - Transparency

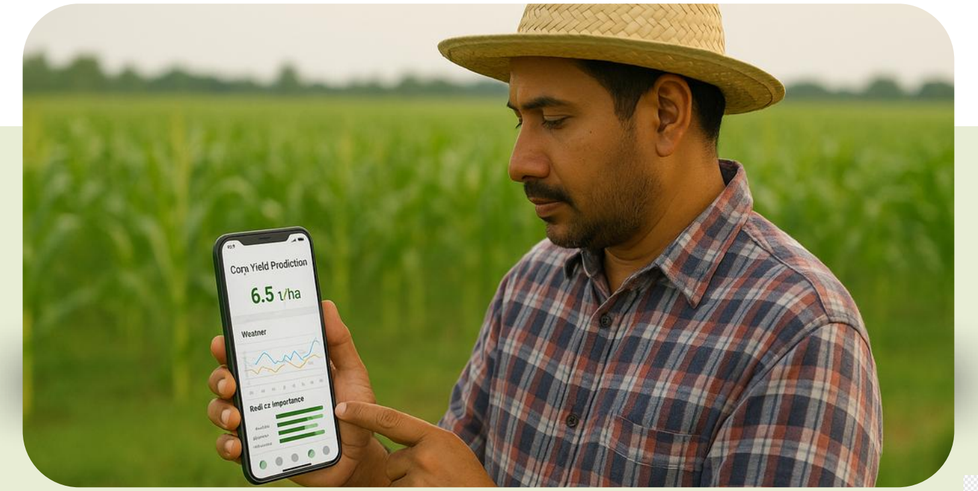
ML-BASED CORN YIELD PREDICTION FOR SRI LANKAN FARMERS

Model Output & Explainability

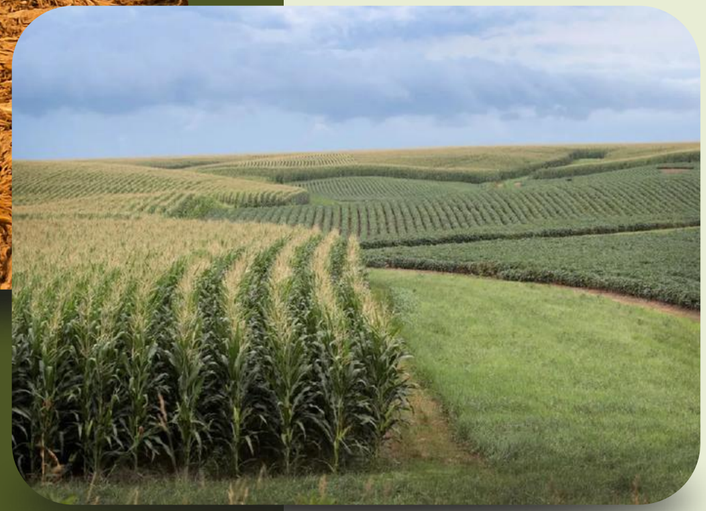
- System predicts yield based on user inputs
- SHAP-based explanation displays:
  - Seasonal rainfall impact
  - Pest/disease level impact
  - Temperature influence
  - Soil type contribution
- Helps users understand why the yield is high or low, not just the number

User Feedback on Prototype

- Users found the mobile interface:
  - Simple and easy to navigate
  - Easy to enter field data
  - Easy to understand prediction results
- Explainable factor breakdown:
  - Improved trust in model predictions
  - Helped users relate results to real farm conditions
- Visual impact indicators made decision-making clearer



ML-BASED CORN YIELD PREDICTION FOR SRI LANKAN FARMERS



Evaluate / Prove Success

- **Technical Evaluation:**
  - Accuracy tested with MAE & R^2 score vs. actual yields.
  - Robustness validated across agro-climatic zones & socioeconomic groups.
- **Usability Evaluation:**
  - Pilot trials with smallholder farmers & extension officers
  - Feedback on clarity, interpretability of SHAP/LIME outputs, and ease of following recommendations..
- **Expected Results:**
  - ≥85% predictive accuracy for yield estimation.
  - Clear explainability outputs showing key yield drivers.
  - High adoption due to simple, farmer-friendly recommendations.

Data Availability & Ethical Clearance

- **Data Sources:**
  - Department of Agriculture (Sri Lanka).
  - Provincial Agricultural Research Stations.
  - Hector Kobbekaduwa Agrarian Research & Training Institute
  - Farmer Organizations & participatory reports.
- **Ethical Considerations:**
  - Informed consent for farmer surveys & data collection.
  - Privacy protection & anonymization of farmer identities.
  - Secure storage & transmission of sensitive data.
  - Compliance with research ethics and local agricultural data governance.

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Project Overview & Commercial Potential

Ability of Commercialization

- Deploy as a mobile app for Sri Lankan maize farmers, agricultural extension officers, and cooperatives.
- Freemium or subscription-based model, with optional premium services like advanced disease analytics, alerts, or integration with knowledge-sharing platforms.
- Addresses a real problem: Corn diseases cause 11–50% yield loss in Sri Lanka, motivating farmers to adopt the tool.

Value Proposition

- Early detection of foliar diseases → reduces yield losses by up to 30–50%.
- Farmer-friendly & offline-capable mobile app → usable in rural areas with limited connectivity.
- Explainable AI → builds trust by showing diseased regions.
- Integration with knowledge-sharing platform → improves disease management and farmer education.

Investment / Cost & Recovery

- Labor Cost: App & AI model development, data collection, UI/UX design.
- Material Cost: Smartphones for testing, cloud storage, servers for initial model training.
- Recovery: Pricing model example:

  Small monthly subscription: ~LKR 400–1000 per farmer/month.

  Institutional licenses for cooperatives or extension programs.

  Partnerships with agritech startups or government subsidy programs.

- Primary customers are small- and medium-scale maize farmers in Sri Lanka
- Secondary customers would be Agricultural advisors, NGOs, extension officers, and agritech companies.
- AI models, preprocessing pipelines, and mobile app interface can be protected under copyright and software patents.

IT4010 Research Project –2025 July Presentation

Presented By: **25-26J-239**

We're excited about building a sustainable future for agriculture.

Thank You Very Much